MATH1073: Calculus I Midterm Solution

1. (9 points) Find the least upper bound and the greatest lower bound of each of the following sets. Or if it is not possible, explain why.

   (a) $\{x : |x-2| \le 3\}$.

   (b) $\{x : x^2 > 3\}$.

   (c) $\{x : \cos x \le 1\}$.

   (d) $\{x : \ln x \le e\}$.

   (e) $\left\{\frac{n-1}{n+1}\cos\left(\frac{2n\pi}{3}\right) : n \in \mathbb{N}\right\}$.

   (f) $\left\{(1+(-1)^n)\cdot\frac{n+1}{n} : n \in \mathbb{N}\right\}$.

   Solution:

   (a) $|x-2| \le 3 \Leftrightarrow -3 \le x-2 \le 3 \Leftrightarrow -1 \le x \le 5$. So $lub = 5, \quad glb = -1$.

   (b) $x^2 > 3 \Leftrightarrow x > \sqrt{3}$ or $x < -\sqrt{3}$. Since
   $$\left\{x : x^2 > 3\right\} = (-\infty, -\sqrt{3}) \cup (\sqrt{3}, \infty)$$
   is not bounded above nor below, the lub and glb does not exist.

   (c) Since $\{x : \cos x \le 1\} = \mathbb{R}$, which is not bounded below nor above, this set does not have lub nor glb.

   (d) $\ln x \le e \Leftrightarrow 0 < x \le e^e$. So $lub = e^e, glb = 0$.

   (e) Note that $0 \le \frac{n-1}{n+1} < 1$ for $n \in \mathbb{N}$ and $\frac{n-1}{n+1} \to 1$ as $n \to \infty$, and $\cos\left(\frac{2n\pi}{3}\right) \in \left\{1, -\frac{1}{2}\right\}$. So $lub = 1, \quad glb = -\frac{1}{2}$.

   (f) Note that $1+(-1)^n = \begin{cases} 0, & \text{if } n \text{ is odd} \\ 2, & \text{if } n \text{ is even} \end{cases}$ and $\frac{n+1}{n}$ is decreasing as n increases. So $lub = 2 \cdot \frac{2+1}{2} = 3, \quad glb = 0$.

2. (10 points)

   (a) Let $a_n = \frac{1}{n+n^2}$ for $n \in \mathbb{N}$. Use the "$\epsilon - N$" definition of a limit to prove that $\lim_{n\to\infty} a_n = 0$.

   (b) Let $a_n = \frac{1}{2+1} + \frac{1}{2^2+1} + \cdots + \frac{1}{2^n+1}$ for $n \in \mathbb{N}$. Show that the limit of $\{a_n\}$ exists.

   Solution:

   (a) Let $\varepsilon > 0$ be given. Note that
   $$|a_n - 0| = \left|\frac{1}{n+n^2} - 0\right| = \frac{1}{n+n^2} < \frac{1}{n}.$$
   We choose $N = \left\lceil \frac{1}{\varepsilon} \right\rceil + 1$. Then for $n \ge N$,
   $$|a_n - 0| < \frac{1}{n} \le \frac{1}{N} = \frac{1}{\left\lceil \frac{1}{\varepsilon} \right\rceil + 1} < \frac{1}{\frac{1}{\varepsilon}} = \varepsilon.$$
   So $\lim_{n\to\infty} a_n = 0$.

(b) Obviously $\{a_n\}$ is increasing. By the monotone convergence theorem, we just need to show that $\{a_n\}$ is bounded above. Note that

$$\begin{aligned} a_n &= \frac{1}{2+1} + \frac{1}{2^2+1} + \cdots + \frac{1}{2^n+1} \\ &\le \frac{1}{2} + \frac{1}{2^2} + \cdots + \frac{1}{2^n} \\ &= \frac{\frac{1}{2} \cdot \left(1 - \left(\frac{1}{2}\right)^n\right)}{1 - \frac{1}{2}} \le \frac{\frac{1}{2}}{1 - \frac{1}{2}} = 1. \end{aligned}$$

So $\{a_n\}$ is bounded and $\{a_n\}$ converges.

3. (18 points) Determine whether the sequence converges or diverges. If it is convergent, find the limit.

(a) $a_n = \frac{3n^3+n^2-2n}{4n^3+n-1}$.

(b) $a_n = \frac{100n+1}{(2n-1)^2}$.

(c) $a_n = \frac{\sin(n!)}{\sqrt{n}}$.

(d) $a_n = \frac{1+2+3+\cdots+n}{n+2} - \frac{n}{2}$.

(e) $a_n = \sqrt{n+2} - \sqrt{n}$.

(f) $a_n = \left(1 - \frac{1}{n}\right)^{n^2}$.

Solution:

(a) $\{a_n\}$ converges, and

$$\begin{aligned} \lim_{n\to\infty} a_n &= \lim_{n\to\infty} \frac{\left(3n^3+n^2-2n\right)/n^3}{\left(4n^3+n-1\right)/n^3} \\ &= \lim_{n\to\infty} \frac{3 + \frac{1}{n} - \frac{2}{n^2}}{4 + \frac{1}{n^2} - \frac{1}{n^3}} = \frac{3}{4}. \end{aligned}$$

(b) $\{a_n\}$ converges, and

$$\begin{aligned} \lim_{n\to\infty} a_n &= \lim_{n\to\infty} \frac{(100n+1)/n^2}{(2n-1)^2/n^2} \\ &= \lim_{n\to\infty} \frac{\frac{100}{n} + \frac{1}{n^2}}{\left(2 - \frac{1}{n}\right)^2} = \frac{0}{2^2} = 0. \end{aligned}$$

(c) Note that $-\frac{1}{\sqrt{n}} \le \frac{\sin(n!)}{\sqrt{n}} \le \frac{1}{\sqrt{n}}$. By the squeeze theorem,

$$\lim_{n\to\infty} \frac{\sin(n!)}{\sqrt{n}} = 0.$$

(d) Note that

$$\begin{aligned} a_n &= \frac{1+2+\cdots+n}{n+2} - \frac{n}{2} \\ &= \frac{n(n+1)}{2(n+2)} - \frac{n}{2} \\ &= \frac{n}{2} \cdot \left(\frac{n+1}{n+2} - 1\right) = \frac{n}{2} \cdot \frac{-1}{n+2}. \end{aligned}$$

So

$$\begin{aligned}\lim_{n\to\infty} a_n &= \lim_{n\to\infty} -\frac{1}{2}\cdot\frac{n}{n+2}\\ &= \lim_{n\to\infty} -\frac{1}{2}\cdot\frac{1}{1+\frac{2}{n}} = -\frac{1}{2}.\end{aligned}$$

(e) It holds that

$$\begin{aligned}\lim_{n\to\infty}(\sqrt{n+2}-\sqrt{n}) &= \lim_{n\to\infty}\frac{(\sqrt{n+2}-\sqrt{n})(\sqrt{n+2}+\sqrt{n})}{\sqrt{n+2}+\sqrt{n}}\\ &= \lim_{n\to\infty}\frac{2}{\sqrt{n+2}+\sqrt{n}}\\ &= \lim_{n\to\infty}\frac{2/\sqrt{n}}{\sqrt{1+\frac{2}{n}}+1}\\ &= \frac{0}{2} = 0.\end{aligned}$$

(f) Since

$$\begin{aligned}\left(1-\frac{1}{n}\right)^{n^2} &= \left[\left(1-\frac{1}{n}\right)^{-1}\right]^{-n^2}\\ &= \left[\frac{n}{n-1}\right]^{-n^2}\\ &= \left[\left(1+\frac{1}{n-1}\right)^{n-1}\right]^{-\frac{n^2}{n-1}}\end{aligned}$$

and $\left(1+\frac{1}{n-1}\right)^{n-1}\to e$ as $n\to\infty$ and $-\frac{n^2}{n-1}\to-\infty$ as $n\to\infty$, we have

$$\lim_{n\to\infty} a_n = 0.$$

4. (18 points) Find the limit if it exists. If the limit does not exist, explain why.

(a) $\lim_{x\to\infty}\frac{x+1}{x}$.

(b) $\lim_{x\to 1}\frac{x-1}{x^2-1}$.

(c) $\lim_{x\to 2^+}\frac{\cos(\pi x)}{8-x^3}$.

(d) $\lim_{x\to\infty}\frac{x+1}{\sqrt{x^2+1}}$.

(e) $\lim_{x\to -1}\frac{4x^2+7x}{x^3+1}$.

(f) $\lim_{x\to 0} x\cot x$.

Solution:

(a) $\lim_{x\to\infty}\frac{x+1}{x} = \lim_{x\to\infty}\frac{1+\frac{1}{x}}{1} = 1$.

(b) $\lim_{x\to 1}\frac{x-1}{x^2-1} = \lim_{x\to 1}\frac{x-1}{(x+1)(x-1)} = \lim_{x\to 1}\frac{1}{x+1} = \frac{1}{2}$.

(c) Note that if $x > 2$, then $8 - x^3 < 0$. Moreover, as $x \to 2^+$, we have $8 - x^3 \to 0$. So $\lim_{x\to 2^+} \frac{\cos(\pi x)}{8-x^3} = -\infty$.

(d) It holds that

$$\lim_{x\to\infty} \frac{x+1}{\sqrt{x^2+1}} = \lim_{x\to\infty} \frac{(x+1)/x}{\sqrt{x^2+1}/x}$$
$$= \lim_{x\to\infty} \frac{1+\frac{1}{x}}{\sqrt{1+\frac{1}{x^2}}} = \frac{1}{1} = 1.$$

(e) Note that $4x^2 + 7x \to -3$ and $x^3 + 1 \to 0$ as $x \to -1$. Moreover, $x^3 + 1 = (x+1)(x^2 - x + 1)$

$$= (x+1)\left[\left(x - \frac{1}{2}\right)^2 + \frac{3}{4}\right],$$

So $x^3 + 1 > 0$ if $x > -1$ and $x^3 + 1 < 0$ if $x < -1$. As a result, $\lim_{x\to -1} \frac{4x^2+7x}{x^3+1}$ does not exist, since

$$\lim_{x\to -1^+} \frac{4x^2+7x}{x^3+1} = -\infty \text{ and } \lim_{x\to 1^-} \frac{4x^2+7x}{x^3+1} = \infty.$$

(f) It holds that

$$\lim_{x\to 0} x\cot x = \lim_{x\to 0} x \cdot \frac{\cos x}{\sin x} = \lim_{x\to 0} \cos x \cdot \frac{x}{\sin x}$$
$$= \lim_{x\to 0} \cos x \cdot \frac{1}{\frac{\sin x}{x}}$$
$$= 1 \cdot 1 = 1.$$

5. (8 points) Prove that if g is continuous at a and f is continuous at $g(a)$, then the composite function $f \circ g$ given by $(f \circ g)(x) = f(g(x))$ is continuous at a.

Proof: Let $\epsilon > 0$ be given. We want to find a number $\delta > 0$ such that
if $0 < |x - a| < \delta$ then $|f(g(x)) - f(g(a))| < \epsilon$
Since f is continuous at $b = g(a)$, we have

$$\lim_{y\to b} f(y) = f(b)$$

and so there exits $\delta_1 > 0$ such that
if $0 < |y - b| < \delta_1$ then $|f(y) - f(b)| < \epsilon$
Since g is continuous at a, we have

$$\lim_{x\to a} g(x) = g(a)$$

and so there exits $\delta > 0$ such that
if $0 < |x - a| < \delta$ then $|g(x) - b| < \delta_1$
Combining these two statements, we see that whenever $0 < |x - a| < \delta$ we have $|g(x) - b| < \delta_1$, which implies that $|f(g(x)) - f(b)| < \epsilon$. Therefore we have proved that

$$\lim_{x\to a} f(g(x)) = f(g(a))$$

which is precisely the statement that the function $f(g(x))$ is continuous at a; that is, $f \circ g$ is continuous at a.

6. (8 points) Show that the function

$$f(x) = \begin{cases} x^4 \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

is continuous on $(-\infty, \infty)$.

Solution: We know that the function x^4 is continuous on $(-\infty, \infty)$, $\sin(1/x)$ is continuous except $x = 0$. So $f(x)$ is continuous on the intervals $(-\infty, 0)$ and $(0, \infty)$.

Here $f(0) = 0$ is defined. Because $|x^4 \sin(1/x)| \leq |x^4| \to 0$, so

$$\lim_{x \to 0} x^4 \sin(\frac{1}{x}) = 0 = f(0).$$

Then f is continuous at 0. Therefore, f is continuous everywhere.

7. (6 points) If a and b are positive numbers, prove that the euqation

$$\frac{a}{x^3 + 2x^2 - 1} + \frac{b}{x^3 + x - 2} = 0$$

has at least one solution in the interval $(-1, 1)$.

Solution: Let $f(x) = a(x^3 + x - 2) + b(x^3 + 2x^2 - 1)$. We are looking for a solution of the given equation, that is, a number c between -1 and 1 such that $f(c) = 0$. We have

$$f(-1) = -4a < 0 \text{ and } f(1) = 2b > 0 \text{ because } a > 0, b > 0$$

The function f is continuous since it is a polynomial, so the Intermediate Value Theorem says there is a number c between -1 and 1 such that $f(c) = 0$.

8. (15 points) Calculate y'.

   (a) $y = \left(\frac{x-1}{x^2+x+1}\right)^4$
   (b) $y = x^2 \sin \pi x$
   (c) $y = (\cos x)^x$
   (d) $y = (\arcsin 2x)^2$
   (e) $xe^y = y - 1$

Solution:

(a) Since

$$y = \left(\frac{x-1}{x^2+x+1}\right)^4,$$

we have

$$\begin{aligned} y' &= 4\left(\frac{x-1}{x^2+x+1}\right)^3 \frac{d}{dx}\left(\frac{x-1}{x^2+x+1}\right) \\ &= 4\left(\frac{x-1}{x^2+x+1}\right)^3 \frac{(x^2+x+1)(1) - (x-1)(2x+1)}{(x^2+x+1)^2} \\ &= \frac{4(x-1)^3}{(x^2+x+1)^3} \frac{x^2+x+1-2x^2+x+1}{(x^2+x+1)^2} = \frac{4(x-1)^3(-x^2+2x+2)}{(x^2+x+1)^5} \end{aligned}$$

(b) $y = x^2 \sin \pi x \quad \Rightarrow \quad y' = x^2(\cos \pi x)\pi + (\sin \pi x)(2x) = x(\pi x \cos \pi x + 2 \sin \pi x)$

(c) $y = (\cos x)^x \Rightarrow \ln y = \ln(\cos x)^x = x \ln \cos x \Rightarrow \dfrac{y'}{y} = x \cdot \dfrac{1}{\cos x} \cdot (-\sin x) + \ln \cos x \cdot 1 \Rightarrow$ $y' = (\cos x)^x(\ln \cos x - x \tan x)$

(d) $y = (\arcsin 2x)^2 \Rightarrow y' = 2(\arcsin 2x) \cdot (\arcsin 2x)' = 2 \arcsin 2x \cdot \frac{1}{\sqrt{1-(2x)^2}} \cdot 2 = \frac{4 \arcsin 2x}{\sqrt{1-4x^2}}$

(e) $xe^y = y-1 \quad \Rightarrow \quad xe^y y' + e^y = y' \quad \Rightarrow \quad e^y = y' - xe^y y' \quad \Rightarrow \quad y' = e^y / (1 - xe^y)$

9. (8 points) Find the linearization of $f(x) = \sqrt[3]{1+3x}$ at $x = 0$ and use it to give an approximate value for $\sqrt[3]{1.03}$.

   Solution: $f(x) = \sqrt[3]{1+3x} = (1+3x)^{1/3} \Rightarrow f'(x) = (1+3x)^{-2/3}$, so the linearization of f at $x = 0$ is $L(x) = f(0) + f'(0)(x-0) = 1^{1/3} + 1^{-2/3}x = 1 + x$. Thus, $\sqrt[3]{1+3x} \approx 1 + x \Rightarrow \sqrt[3]{1.03} = \sqrt[3]{1+3(0.01)} \approx 1 + (0.01) = 1.01$

10. (**Bonus question**, 8 points) Show that the equation $x^3 - 15x + c = 0$ has at most one root in the interval $[-2, 2]$, where $c \in \mathbb{R}$ is a constant.

    Proof:

    Define

    $$f(x) = x^3 - 15x + c.$$

    Suppose that the equation has at least two roots in the interval $[-2, 2]$. As a result, there are x_1, x_2 with $-2 \leq x_1 < x_2 \leq 2$ such that

    $$f(x_1) = f(x_2) = 0.$$

    By Rolle's theorem, there must be $c \in (x_1, x_2)$ such that $f'(c) = 0$. However, since $-2 \leq c \leq 2$, we have

    $$f'(c) = 3c^2 - 15 \leq 3 \cdot 2^2 - 15 = -3,$$

    a contradiction. Therefore, our assumption that the equation has at least two roots in the interval $[-2, 2]$ was false. The equation $x^3 - 15x + c = 0$ has at most one root in the interval $[-2, 2]$.